

Lecture 1

INTRODUCTION & RECTILINEAR KINEMATICS: CONTINUOUS MOTION

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An Overview of Mechanics

Mechanics: The study of how bodies react to the forces acting on them.

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graph TD; A[Mechanics: The study of how bodies react to the forces acting on them.] --> B[Statics: The study of bodies in equilibrium.]; A --> C[Dynamics: The study of bodies in motion.];
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Statics: The study of bodies in equilibrium.

Dynamics: The study of bodies in motion.

INTRODUCTION

Dynamics includes:

Kinematics: study of the geometry of motion.

Relates displacement, velocity, acceleration, and time *without reference* to the cause of motion.

Kinetics: study of the relations existing between the forces acting on a body, the mass of the body, and the motion of the body.

Predict the motion caused by given forces or to determine the forces required to produce a given motion.

INTRODUCTION

Kinematics of Particles

- ***Rectilinear motion:*** position, velocity, and acceleration of a particle as it moves along a straight line.

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- ***Curvilinear motion:*** position, velocity, and acceleration of a particle as it moves along a curved line in two or three dimensions.

INTRODUCTION

Module 3 (Kinematics of Particles)

1. Rectilinear Motion : Continuous Motion
2. Rectilinear Motion : Erratic Motion
3. Curvilinear Motion : Rectangular Components
4. Curvilinear Motion : Normal & Tangential Components
5. Curvilinear Motion : Cylindrical Components

INTRODUCTION & RECTILINEAR KINEMATICS: CONTINUOUS MOTION

Today's Objectives:

Students will be able to:

1. Find the kinematic quantities (position, displacement, velocity, and acceleration) of a particle traveling along a straight path.



In-Class Activities:

- Reading Quiz
- Relations between $s(t)$, $v(t)$, and $a(t)$ for general rectilinear motion.
- Relations between $s(t)$, $v(t)$, and $a(t)$ when acceleration is constant.
- Example Problem
- Concept Quiz
- Group Problem Solving
- Attention Quiz

READING QUIZ

1. In dynamics, a particle is assumed to have _____.
A) both translation and rotational motions
B) only a mass
C) a mass but the size and shape cannot be neglected
D) no mass or size or shape, it is just a point
2. The average speed is defined as _____.
A) $\Delta r / \Delta t$ B) $\Delta s / \Delta t$
C) $s_T / \Delta t$ D) None of the above.

APPLICATIONS

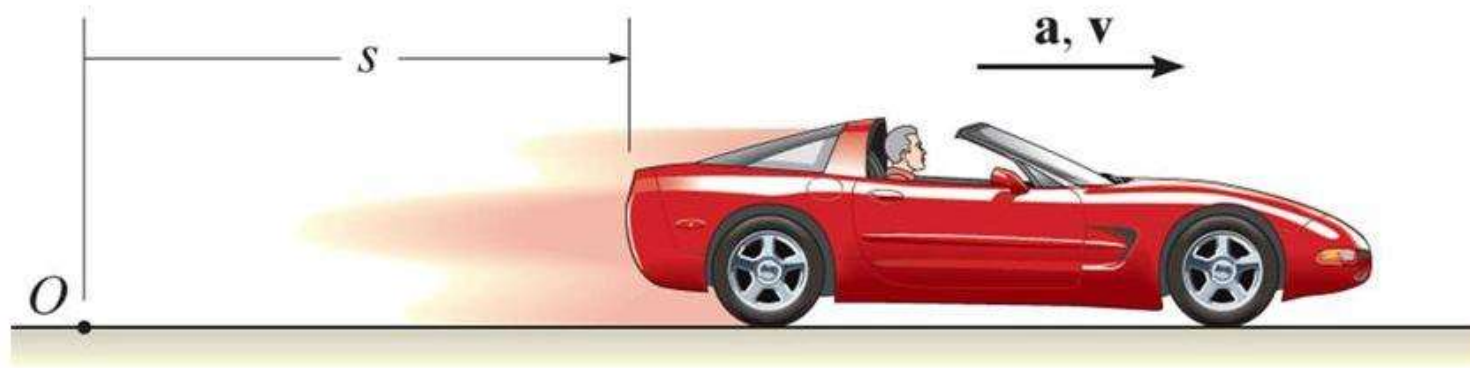


The motion of large objects, such as rockets, airplanes, or cars, can often be analyzed as if they were particles.

Why?

If we measure the altitude of this rocket as a function of time, how can we determine its velocity and acceleration?

APPLICATIONS (continued)

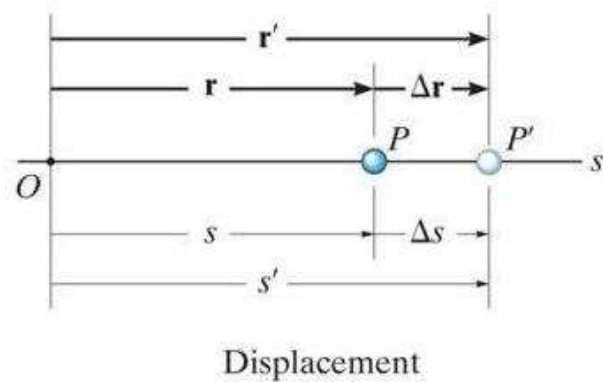
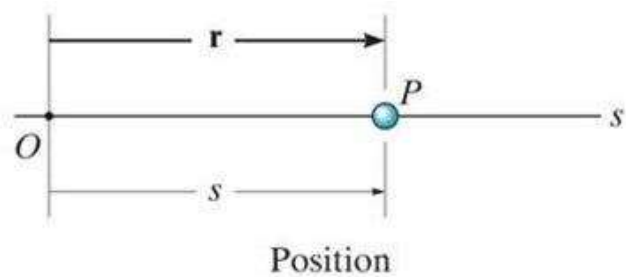


A sports car travels along a straight road.

Can we treat the car as a particle?

If the car accelerates at a constant rate, how can we determine its position and velocity at some instant?

RECTILINEAR KINEMATICS: CONTINUOUS MOTION (Section 12.2)



A particle travels along a straight-line path defined by the **coordinate axis** s .

The **position** of the particle at any instant, relative to the origin, O , is defined by the position vector $\mathbf{r}$, or the scalar s . Scalar s can be positive or negative. Typical units for $\mathbf{r}$ and s are meters (m).

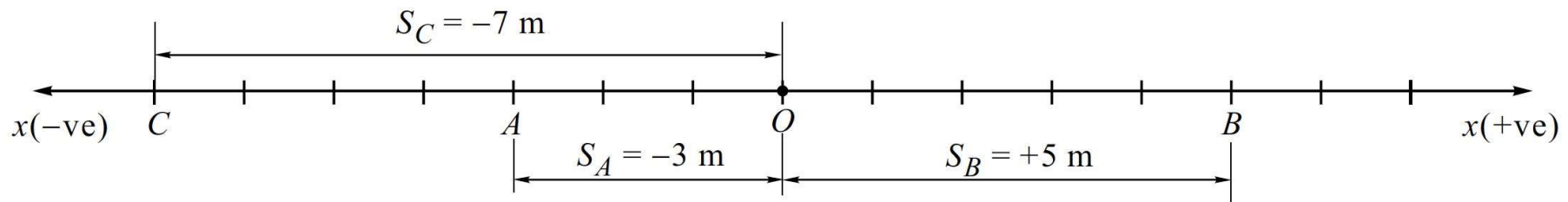
The **displacement** of the particle is defined as its change in position.

Vector form: $\Delta \mathbf{r} = \mathbf{r}' - \mathbf{r}$

Scalar form: $\Delta s = s' - s$

The **total distance traveled** by the particle, s_T , is a positive scalar that represents the total length of the path over which the particle travels.

EXAMPLE



Displacement is a change in the position of a particle.

$$S = S_C - S_A$$

$$S = -7 - (-3)$$

$$S = -4 \text{ m}$$

$$S = 4 \text{ m } (\leftarrow)$$

Distance is the total path traveled by a particle from the initial position to the final position

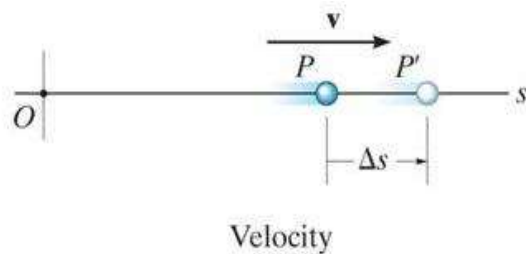
$$d = AO + OB + BO + OC$$

$$d = 3 + 5 + 5 + 7$$

$$d = 20 \text{ m}$$

VELOCITY

Velocity is a measure of the rate of change in the position of a particle. It is a **vector** quantity (it has **both** magnitude and direction). The magnitude of the velocity is called speed, with units of m/s.

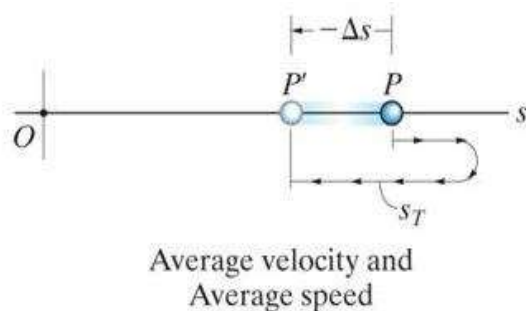


The **average velocity** of a particle during a time interval Δt is

$$\mathbf{v}_{avg} = \Delta \mathbf{s} / \Delta t$$

The **instantaneous velocity** is the time-derivative of position.

$$\mathbf{v} = d\mathbf{s} / dt$$



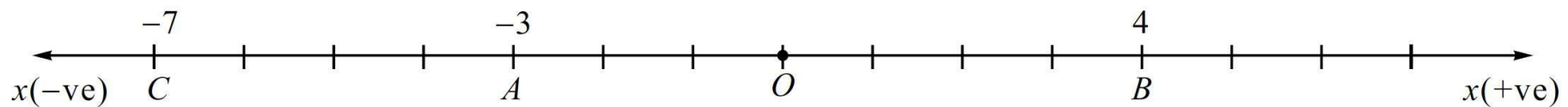
Speed is the magnitude of velocity: $v = ds / dt$

Average speed is the total distance traveled divided by elapsed time:

$$(v_{sp})_{avg} = s_T / \Delta t$$

EXAMPLE

Consider a particle moving from A to B in time $t_1 = 4\text{s}$ and then from B to C in time $t_2 = 6\text{s}$. Find the average velocity and the average speed.



$$\text{Average velocity} = \frac{\text{Change in displacement}}{\text{Time interval}} = \frac{-7 - (-3)}{4 + 6} = \frac{-4}{10}$$

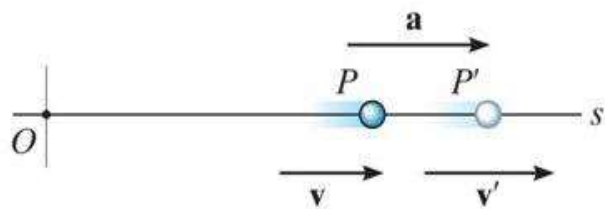
$$\text{Average velocity} = 0.4 \text{ m/s } (\leftarrow)$$

$$\text{Average speed} = \frac{\text{Distance travelled}}{\text{Time interval}} = \frac{3 + 4 + 4 + 7}{4 + 6} = \frac{20}{10}$$

$$\text{Average speed} = 2 \text{ m/s}$$

ACCELERATION

Acceleration is the rate of change in the velocity of a particle. It is a **vector** quantity. Typical units are m/s^2 .

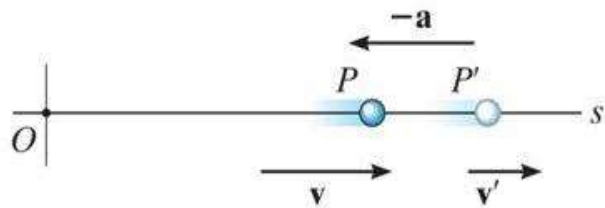


Acceleration

The **instantaneous acceleration** is the time derivative of velocity.

Vector form: $\mathbf{a} = d\mathbf{v} / dt$

Scalar form: $a = dv / dt = d^2s / dt^2$



Deceleration

Acceleration can be positive (speed increasing) or negative (speed decreasing).

As the text shows, the derivative equations for velocity and acceleration can be manipulated to get $a \, ds = v \, dv$

SUMMARY OF KINEMATIC RELATIONS: RECTILINEAR MOTION

- **Differentiate** position to get velocity and acceleration.

$$v = ds/dt ; \quad a = dv/dt \quad \text{or} \quad a = v \, dv/ds$$

- **Integrate** acceleration for velocity and position.

Velocity:

$$\int_{v_o}^v dv = \int_{o}^t a \, dt \quad \text{or} \quad \int_{v_o}^v v \, dv = \int_{s_o}^s a \, ds$$

Position:

$$\int_{s_o}^s ds = \int_{o}^t v \, dt$$

- Note that s_o and v_o represent the **initial position** and **velocity** of the particle at $t = 0$.

CONSTANT ACCELERATION

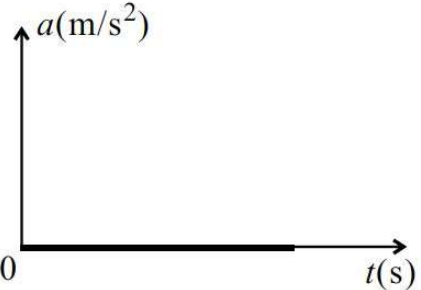
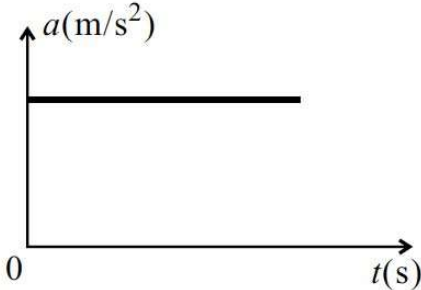
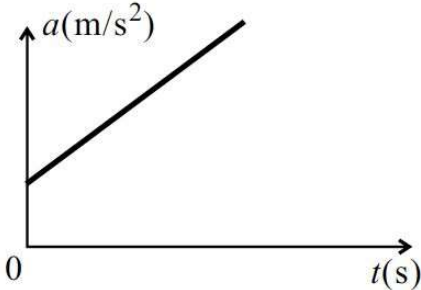
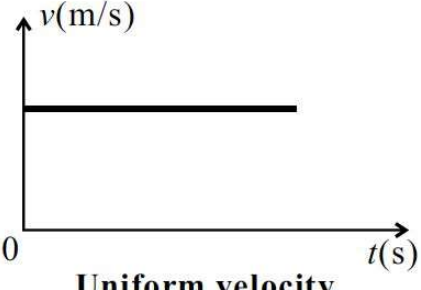
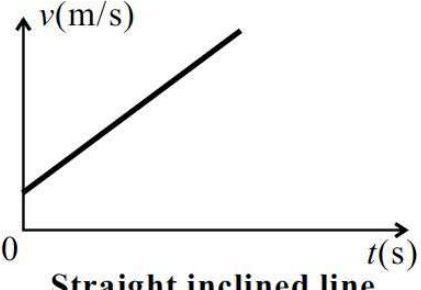
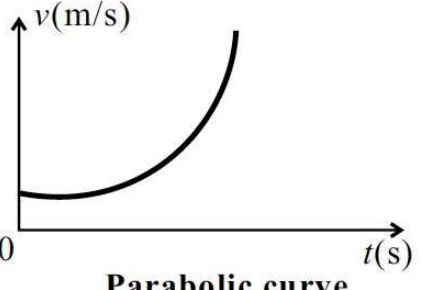
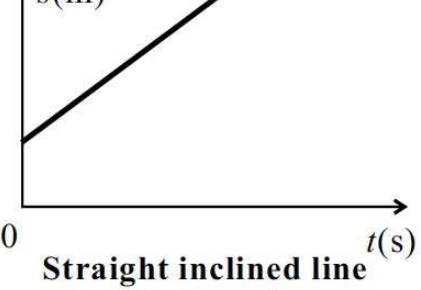
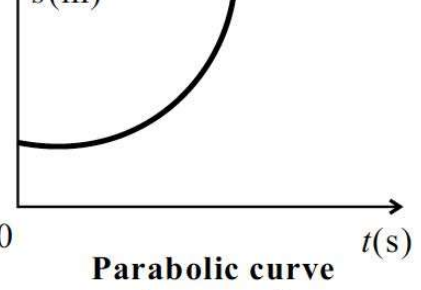
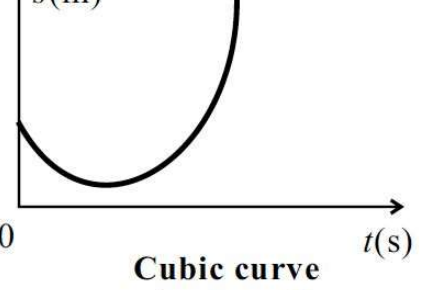
The three kinematic equations can be integrated for the special case when **acceleration is constant** ($a = a_c$) to obtain very useful equations. A common example of constant acceleration is gravity; i.e., a body freely falling toward earth. In this case, $a_c = g = 9.81 \text{ m/s}^2$ downward. These equations are:

$$\int_{v_o}^v dv = \int_0^t a_c dt \quad \text{yield} \quad v = v_o + a_c t$$

$$\int_{s_o}^s ds = \int_0^t v dt \quad \text{yield} \quad s = s_o + v_o t + (1/2) a_c t^2$$

$$\int_{v_o}^v v dv = \int_{s_o}^s a_c ds \quad \text{yield} \quad v^2 = (v_o)^2 + 2a_c (s - s_o)$$

MOTION CURVE

Uniform Velocity Motion Curve	Uniform Acceleration Motion Curve	Variable Acceleration Motion Curve
 Zero acceleration	 Uniform acceleration degree = 0	 Straight inclined line degree = 1
 Uniform velocity	 Straight inclined line degree = 1	 Parabolic curve degree = 2
 Straight inclined line degree = 1	 Parabolic curve degree = 2	 Cubic curve degree = 3

SIGN CONVENTION

Example 1

A ball is thrown vertically up.

Sign Convention

- Initial direction of motion is upwards ($\uparrow$).
- Therefore, upward direction ($\uparrow$) will be positive.
- Velocity in upward direction ($\uparrow$) will be positive.
- Displacement in upward direction ($\uparrow$) will be positive.
- Acceleration due to gravity in upward direction ($\uparrow$) (retardation) will be negative (i.e., $g = -9.81 \text{ m/s}^2$).

Example 2

A ball is thrown vertically down.

Sign Convention

- Initial direction of motion is downwards ($\downarrow$).
- Therefore, downward direction ($\downarrow$) will be positive.
- Velocity in downward direction ($\downarrow$) will be positive.
- Displacement in downward direction ($\downarrow$) will be positive.
- Acceleration due to gravity in downward direction ($\downarrow$) will be positive (i.e., $g = 9.81 \text{ m/s}^2$).

SIGN CONVENTION

Example 3

A car starting from rest moves towards right.

Sign Convention

- Initial direction of motion is towards right ($\rightarrow$).
- Therefore, all kinematic quantities, direction towards right ($\rightarrow$) will be positive but retardation will be negative.

Example 4

A car starting from rest moves towards left.

Sign Convention

- Initial direction of motion is towards left ($\leftarrow$).
- Therefore, all kinematic quantities, direction toward left ($\leftarrow$) will be positive but retardation will be negative.

IMPORTANT POINTS

- Dynamics is concerned with bodies that have accelerated motion.
- Kinematics is a study of the geometry of the motion.
- Kinetics is a study of the forces that cause the motion.
- Rectilinear kinematics refers to straight-line motion.
- Speed refers to the magnitude of velocity.
- Average speed is the total distance traveled divided by the total time. This is different from the average velocity, which is the displacement divided by the time.
- A particle that is slowing down is decelerating.
- A particle can have an acceleration and yet have zero velocity.
- The relationship $a ds = v dv$ is derived from $a = dv/dt$ and $v = ds/dt$, by eliminating dt .

PROCEDURE FOR ANALYSIS

1. Coordinate System

Establish a position coordinate s along the path and specify its fixed origin and positive direction.

The particle's position, velocity, and acceleration can be represented as s , v and a respectively and their sense is then determined from their algebraic signs.

The positive sense for each scalar can be indicated by an arrow shown alongside each kinematics equation as it is applied.

PROCEDURE FOR ANALYSIS

2) Kinematic Equation

If a relationship is known between any two of the four variables a , v , s and t , then a third variable can be obtained by using one of the three the kinematic equations.

When integration is performed, it is important that the position and velocity be known at a given instant to evaluate either the constant of integration if an indefinite integral is used, or the limits of integration if a definite integral is used.

Remember that the three kinematics equations can only be applied to situation where the acceleration of the particle is constant.

EXAMPLE

A particle travels along a straight line to the right with a velocity of $v = (4t - 3t^2)$ m/s where t is in seconds. Determine the acceleration and position of the particle when $t = 4$ s. $s = 0$ when $t = 0$. **(Problem No. F12-3)**

EXAMPLE (continued)

Given: A particle travels along a straight line to the right with a velocity of $v = (4t - 3t^2)$ m/s where t is in seconds. Also, $s = 0$ when $t = 0$.

Find: The position and acceleration of the particle when $t = 4$ s.

Plan: Establish the positive coordinate, s , in the direction the particle is traveling. Since the velocity is given as a **function of time**, take a derivative of it to calculate the acceleration. Conversely, integrate the velocity function to calculate the position.

EXAMPLE (continued)

Solution:

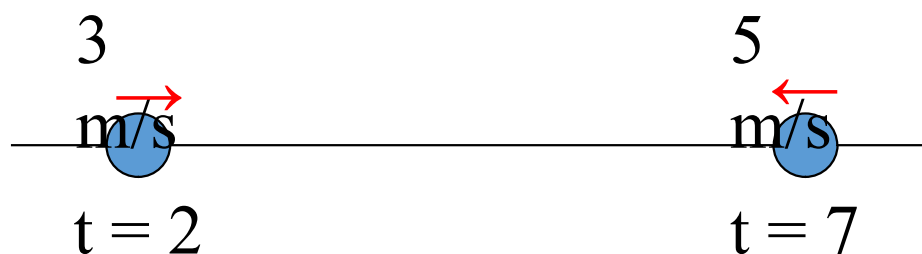
- 1) Take a derivative of the velocity to determine the acceleration.

$$a = dv / dt = d(4t - 3t^2) / dt = 4 - 6t$$
$$\Rightarrow a = \underline{-20 \text{ m/s}^2} \text{ (or in the } \leftarrow \text{ direction) when } t = 4 \text{ s}$$

- 2) Calculate the distance traveled in 4s by integrating the velocity using $s_o = 0$:

$$v = ds / dt \Rightarrow ds = v dt \Rightarrow \int_{s_o}^s ds = \int_0^t (4t - 3t^2) dt$$
$$\Rightarrow s - s_o = 2t^2 - t^3$$
$$\Rightarrow s - 0 = 2(4)^2 - (4)^3 \Rightarrow s = \underline{-32 \text{ m}} \text{ (or } \leftarrow \text{)}$$

CONCEPT QUIZ



1. A particle moves along a horizontal path with its velocity varying with time as shown. The average acceleration of the particle is _____.
 A) $0.4 \text{ m/s}^2 \rightarrow$ B) $0.4 \text{ m/s}^2 \leftarrow$
 C) $1.6 \text{ m/s}^2 \rightarrow$ D) $1.6 \text{ m/s}^2 \leftarrow$
2. A particle has an initial velocity of 30 m/s to the left. If it then passes through the same location 5 seconds later with a velocity of 50 m/s to the right, the average velocity of the particle during the 5 s time interval is _____.
 A) 10 m/s $\rightarrow$ B) 40 m/s $\rightarrow$
 C) 16 m/s $\rightarrow$ D) 0 m/s

GROUP PROBLEM SOLVING

A sandbag is dropped from a balloon which is ascending vertically at a constant speed of 6 m/s. If the bag is released with the same upward velocity of 6 m/s when $t = 0$ and hits the ground when $t = 8$ s, determine the speed of the bag as it hits the ground and the altitude of the balloon at this instant.
(Problem 12-22)

GROUP PROBLEM SOLVING (continued)

Given: A sandbag is dropped from a balloon ascending vertically at a constant speed of 6 m/s. The bag is released with the same upward velocity of 6 m/s at $t = 0$ s and hits the ground when $t = 8$ s.

Find: The speed of the bag as it hits the ground and the altitude of the balloon at this instant.

Plan: The sandbag is experiencing a constant downward acceleration of 9.81 m/s^2 due to gravity. Apply the formulas for constant acceleration, with $a_c = -9.81 \text{ m/s}^2$.

GROUP PROBLEM SOLVING (continued)

Solution:

The bag is released when $t = 0$ s and hits the ground when $t = 8$ s.

Calculate the distance using a position equation.

$$\downarrow + s_{\text{bag}} = (s_{\text{bag}})_o + (v_{\text{bag}})_o t + (1/2) a_c t^2$$

$$s_{\text{bag}} = 0 + (6)(8) + 0.5(-9.81)(8)^2 = 265.9 \text{ m} \downarrow$$

During $t = 8$ s, the balloon rises

$$\uparrow + s_{\text{balloon}} = (v_{\text{balloon}}) t = 6(8) = 48 \text{ m}$$

Therefore, altitude is of the balloon is $(s_{\text{bag}} + s_{\text{balloon}})$.

$$\text{Altitude} = 265.9 + 48 = 313.9 = \underline{314 \text{ m}}.$$

GROUP PROBLEM SOLVING (continued)

Calculate the velocity when $t = 8$ s, by applying a velocity equation.

$$\downarrow + v_{\text{bag}} = (v_{\text{bag}})_o + a_c t$$

$$v_{\text{bag}} = 6 - (9.81) 8 = \underline{72.5 \text{ m/s}} \downarrow$$

ATTENTION QUIZ

1. A particle has an initial velocity of 3 m/s to the left at $s_0 = 0$ m. Determine its position when $t = 3$ s if the acceleration is 2 m/s^2 to the right.

A) 0.0 m B) 6.0 m ←
C) 18.0 m → D) 9.0 m →
2. A particle is moving with an initial velocity of $v = 12 \text{ m/s}$ and constant acceleration of 3.78 m/s^2 in the same direction as the velocity. Determine the distance the particle has traveled when the velocity reaches 30 m/s.

A) 50 m B) 100 m
C) 150 m D) 200 m

End of the Lecture
Let The Learning Continue